

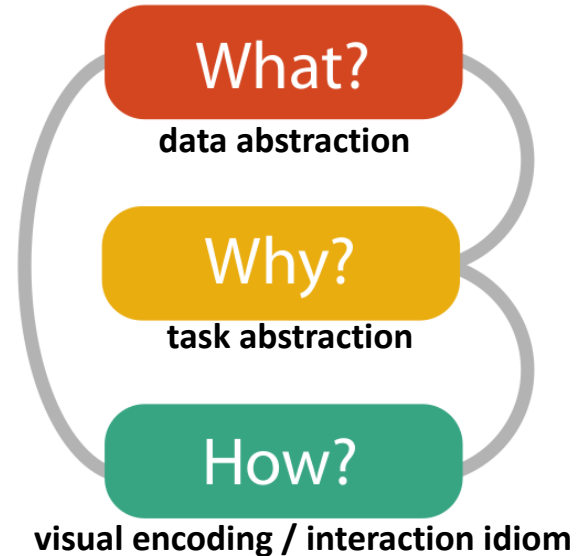
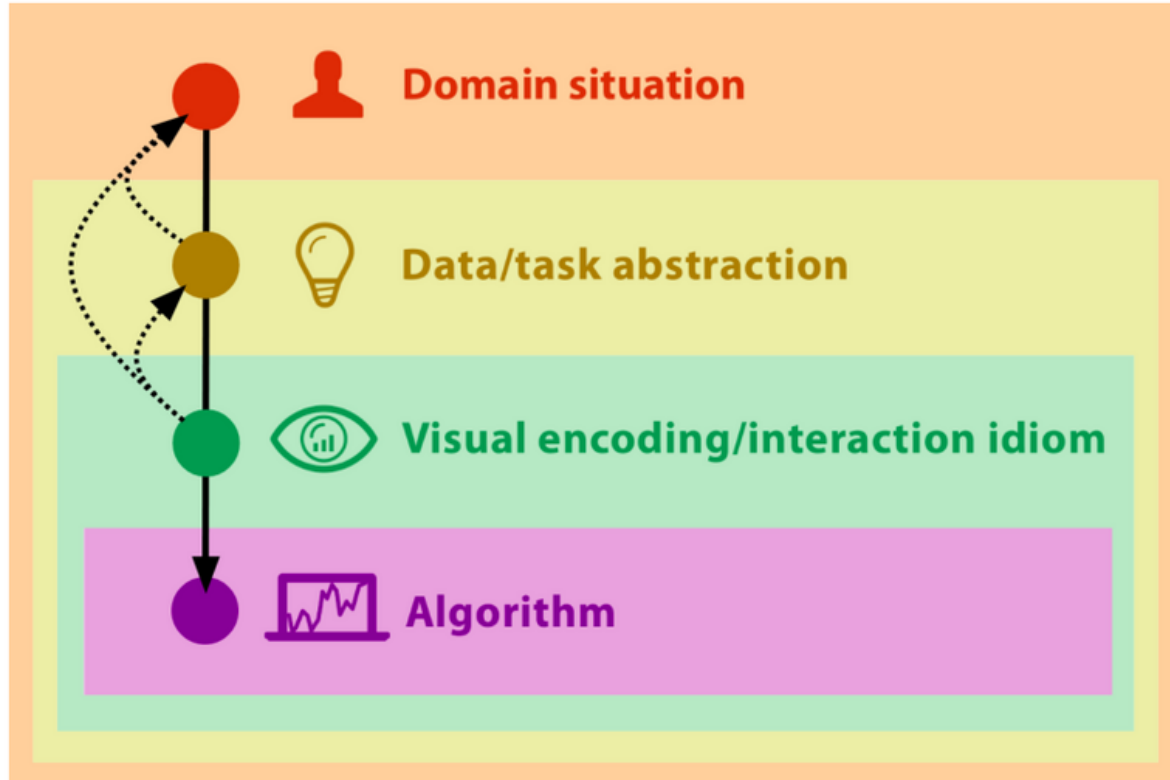
Why? To visualise

DSK808 Data Visualization

Fall 2025

Why?

Participatory Visualization Design



Visual Encoding & Interaction Idiom

How?

Encode

- ➔ **Map**
from **categorical** and **ordered** attributes
 - ➔ Color
 - ➔ Hue
 - ➔ Saturation
 - ➔ Luminance
 - ➔ Size, Angle, Curvature, ...
 -
 - ➔ Shape
 -
 - ➔ Motion
 - Direction, Rate, Frequency, ...*

- ➔ **Arrange**
 - ➔ Express
 - ➔ Order
 - ➔ Use
 - ➔ Separate
 - ➔ Align

Manipulate

- ➔ **Change**
- ➔ **Select**
- ➔ **Navigate**

Facet

- ➔ **Juxtapose**
- ➔ **Partition**
- ➔ **Superimpose**

Reduce

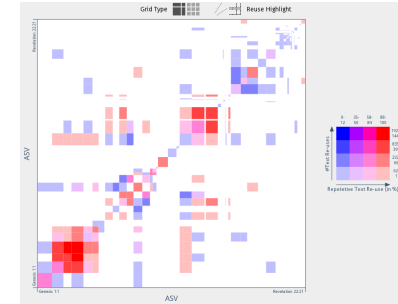
- ➔ **Filter**
- ➔ **Aggregate**
- ➔ **Embed**

Visual Encoding: Arrange

Arrange Tables

Scatterplots, Bar Charts, Line Charts, ...

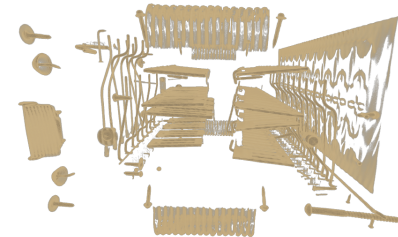
Heat Maps, Timelines, Tag Clouds, ...



Arrange Spatial Data

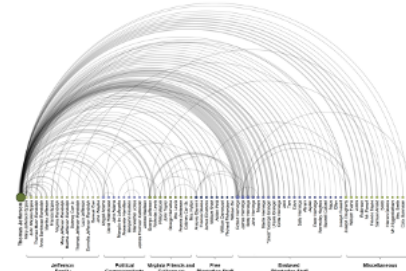
{Choropleth, Bubble, Glyph, ...} Maps

Isokonturen, Direct Volume Rendering (Skalare)

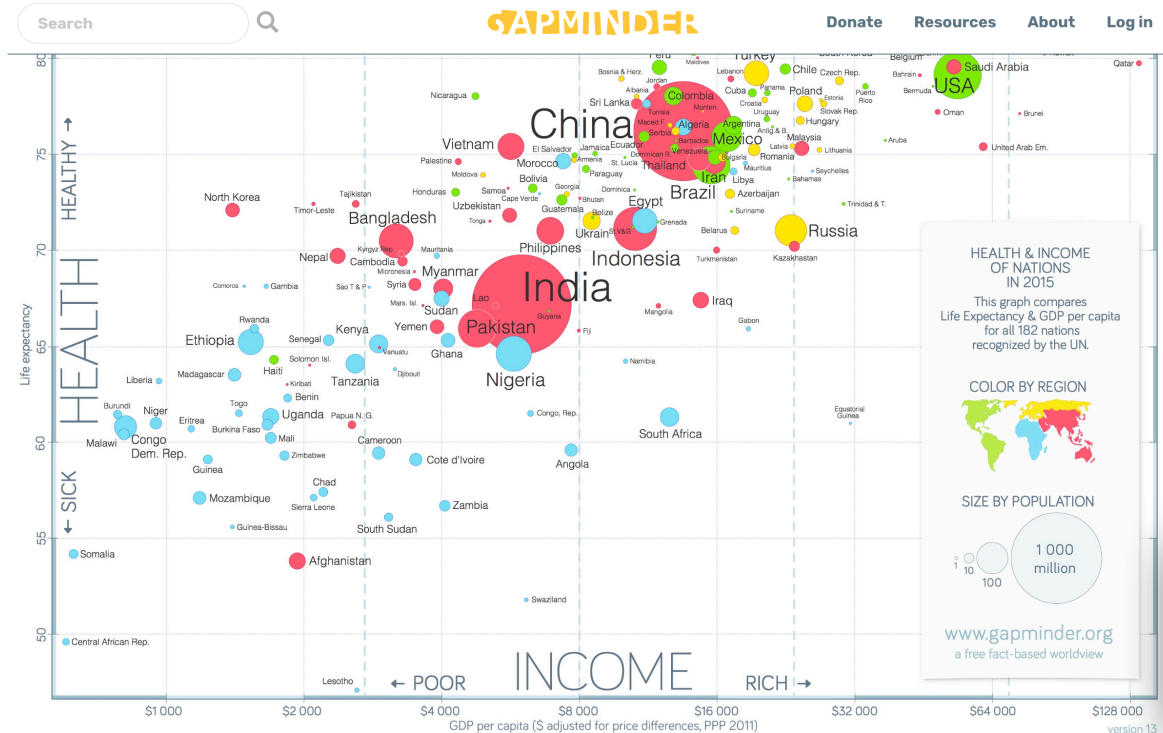


Arrange Networks

Force-Directed Graphs, Radial Layout, TreeMap, ...



Visual Encoding: Arrange

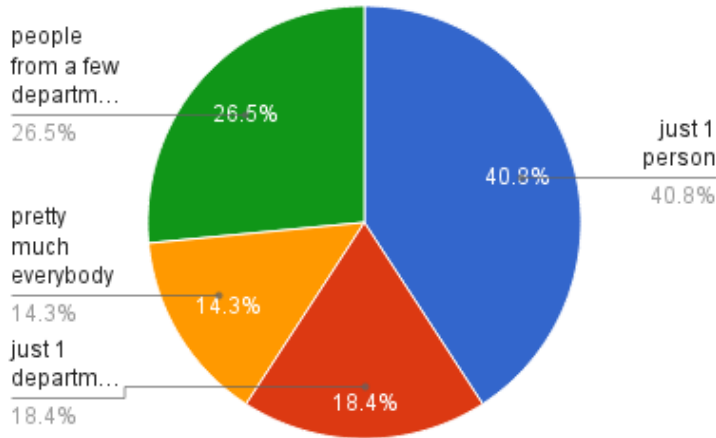


DATA SOURCES—INCOME: World Bank's GDP per capita, PPP (2011 International \$) with a few additions by Gapminder. X-axis uses log scale to make a doubling incomes show same distance on all levels. POPULATION: Numbers from UN Population Division. LIFE EXPECTANCY: IHME GBD-2015, as of Oct 2016. INTERACTIVE GRAPH: www.gapminder.org/tools which lets you animate historic data for hundreds of indicators. LICENSE: Our charts are freely available under Creative Commons Attribution License. Please copy, share, modify, integrate and even sell them, as long as you mention "Based on a free chart from www.gapminder.org".

Idiom	Scatterplots
What: Data	Table: two quantitative value attributes.
How: Encode	Express values with horizontal and vertical spatial position and point marks.
Why: Task	Find trends, outliers, distribution, correlation; locate clusters.
Scale	Items: hundreds.

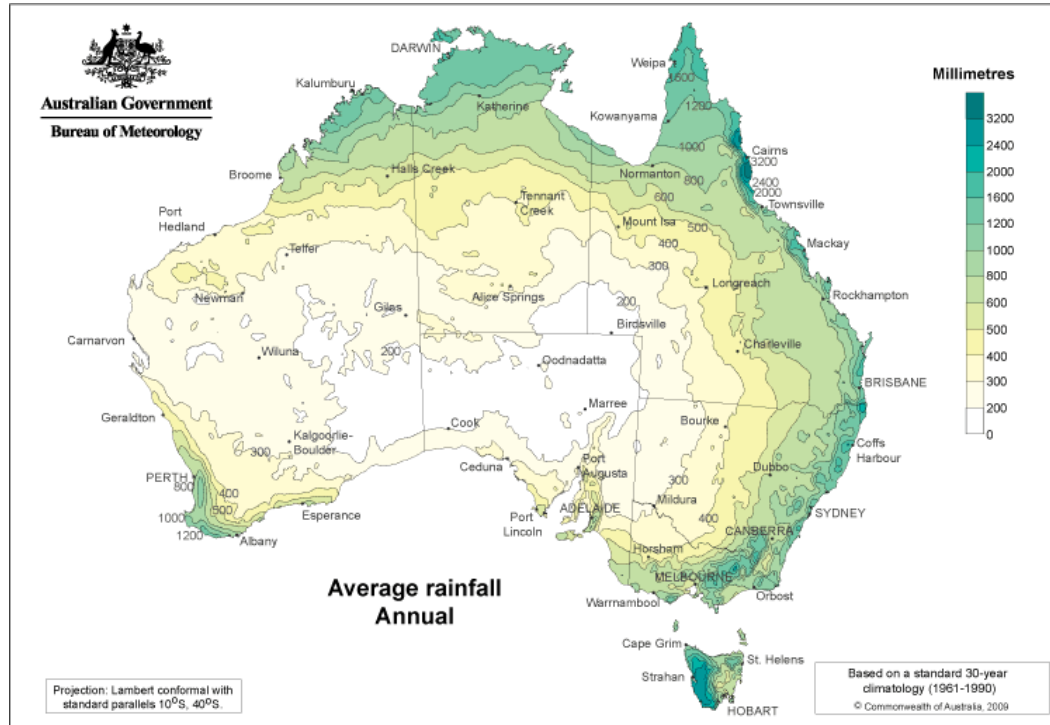
Visual Encoding: Arrange

How many people check web analytics reports in your company?



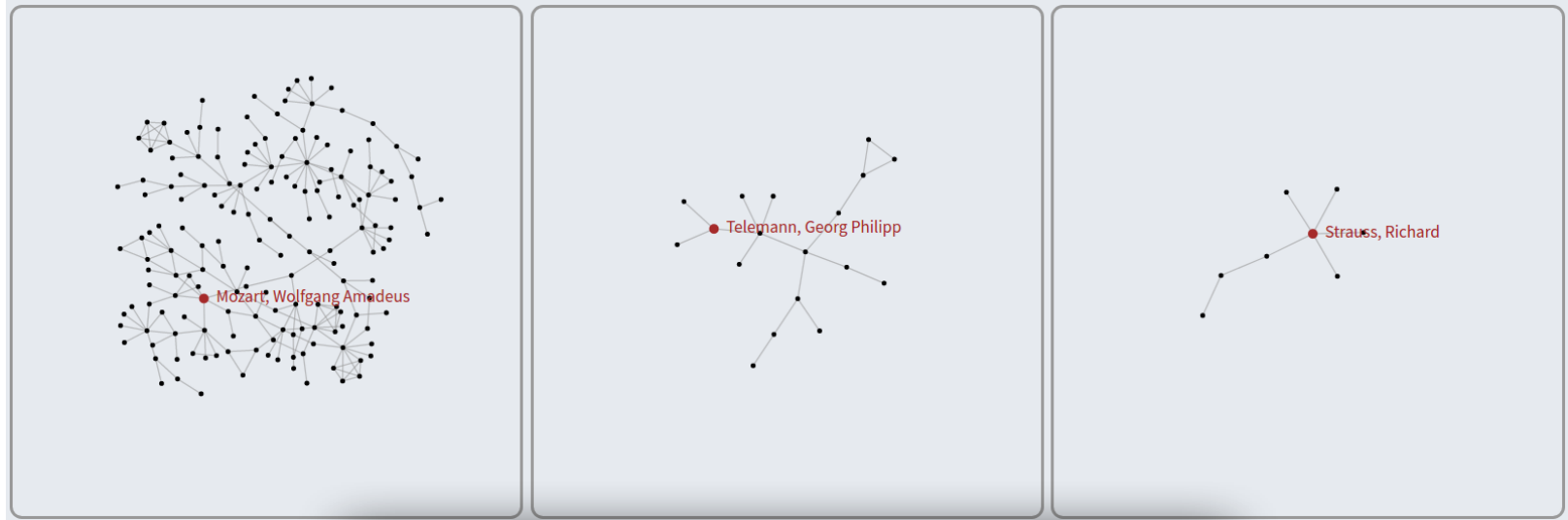
Idiom	Pie Charts
What: Data	Table: one quantitative attribute, one categorical attribute.
Why: Task	Part-whole relationship.
How: Encode	Area marks (wedges) with angle channel; radial layout.
Scale	One dozen categories.

Visual Encoding: Arrange



Idiom	Topographic Terrain Map
What: Data	2D spatial field; geographic data.
What: Derived	Geometry: set of isolines computed from field.
How: Encode	Use given geographic data geometry of points, lines, and region marks. Use derived geometry as line marks (blue).
Why: Tasks	Query shape.
Scale	Dozens of contour levels.

Visual Encoding: Arrange



Idiom	Force-Directed Placement
What: Data	Network.
How: Encode	Point marks for nodes, connection marks for links.
Why: Tasks	Explore topology, locate paths.
Scale	Nodes: dozens/hundreds. Links: hundreds. Node/link density: $L < 4N$

Visual Encoding & Interaction Idiom

How?

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Manipulate

- ➔ **Change**
- ➔ **Select**
- ➔ **Navigate**

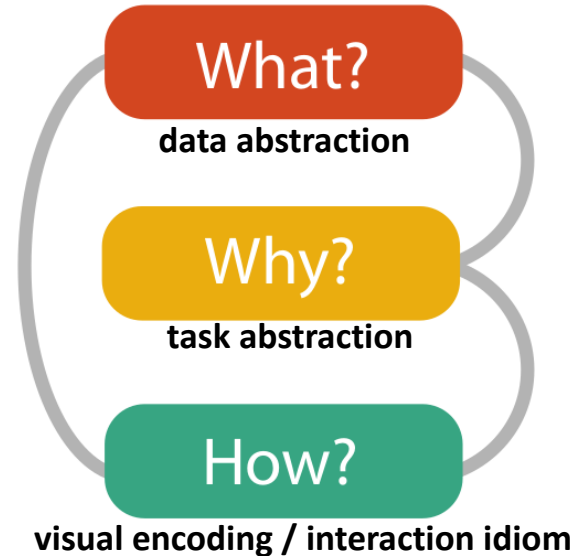
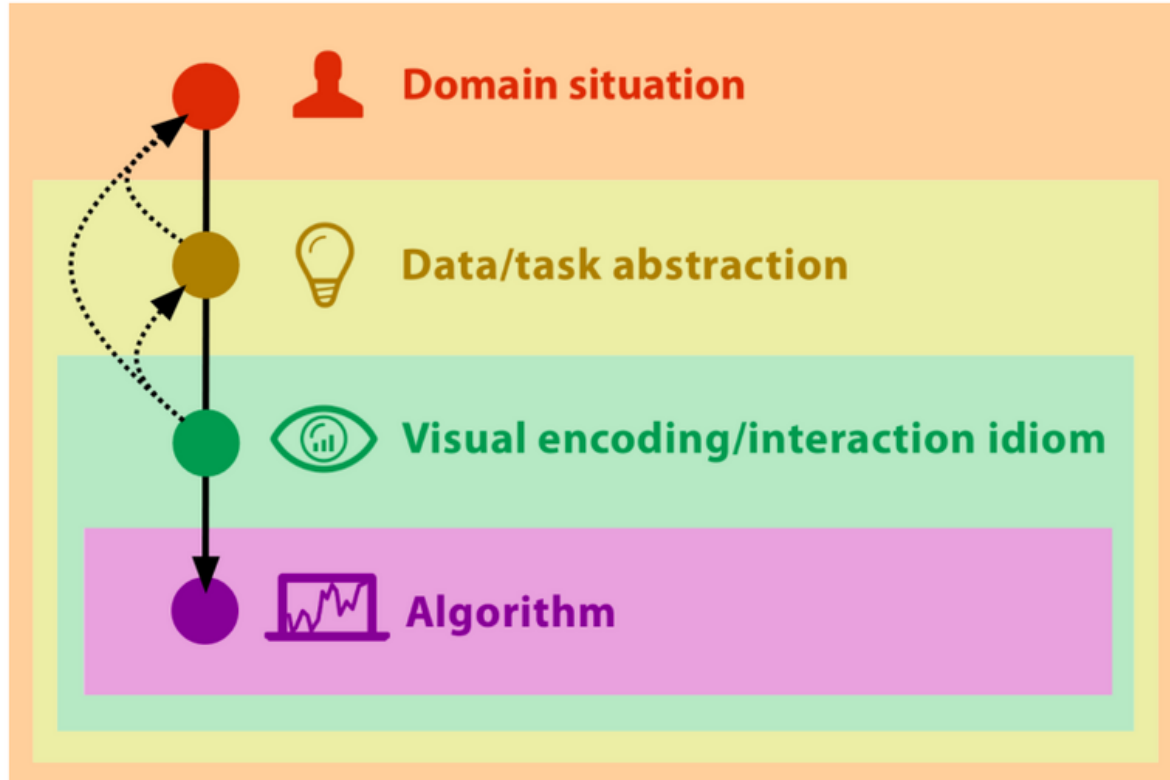
Facet

- ➔ **Juxtapose**
- ➔ **Partition**
- ➔ **Superimpose**

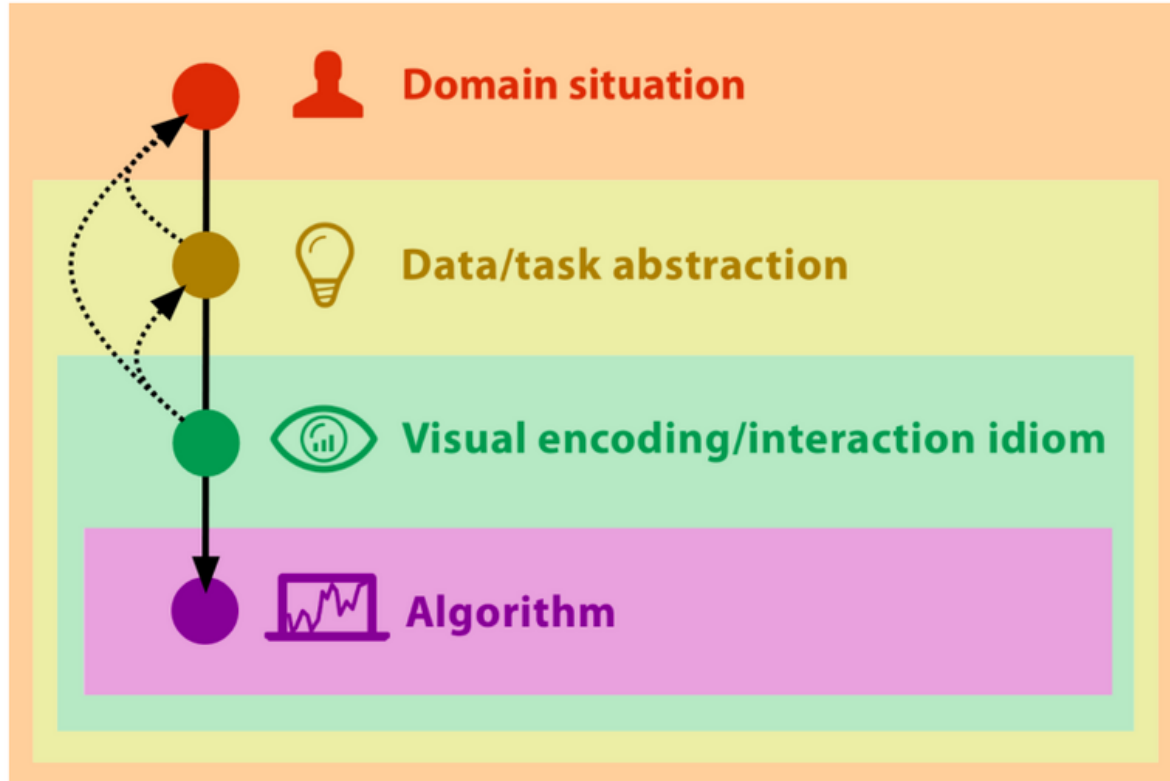
Reduce

- ➔ **Filter**
- ➔ **Aggregate**
- ➔ **Embed**

Participatory Visualization Design

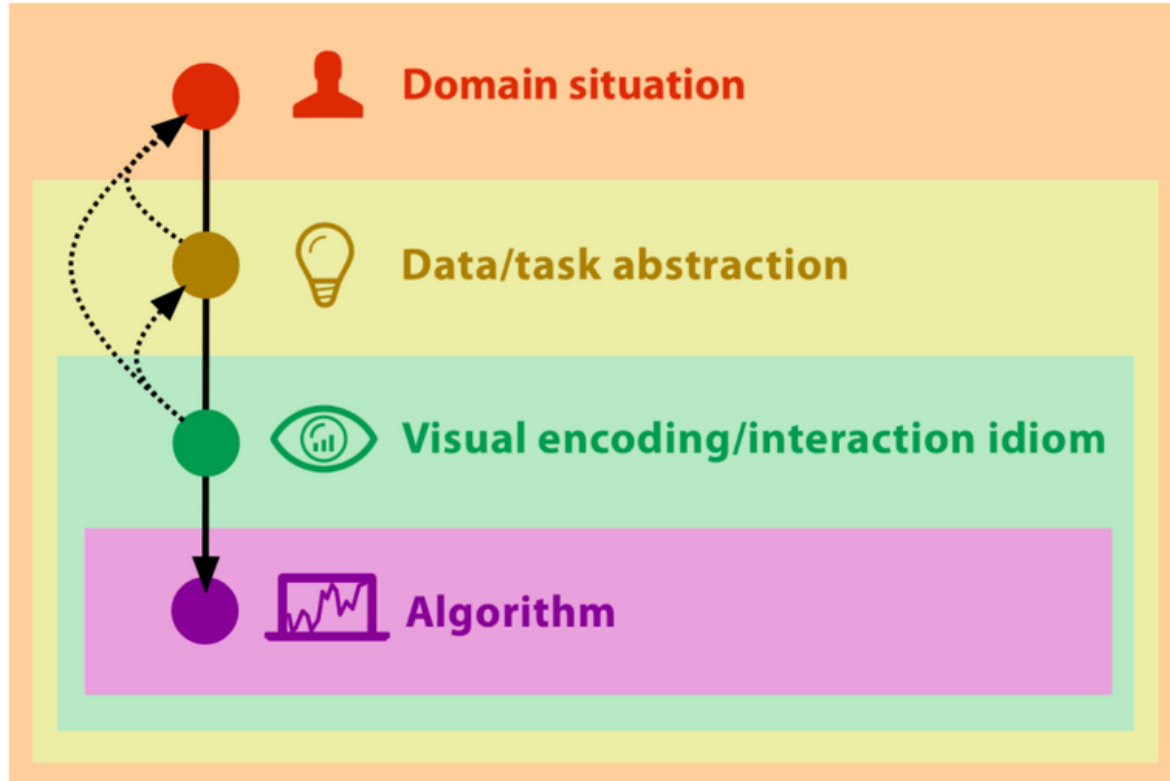


Participatory Visualization Design



- problem-driven development**
- visualization design motivated by **domain-specific questions**
 - often existing tools are sufficient
 - challenge is an accurate abstraction of data and tasks

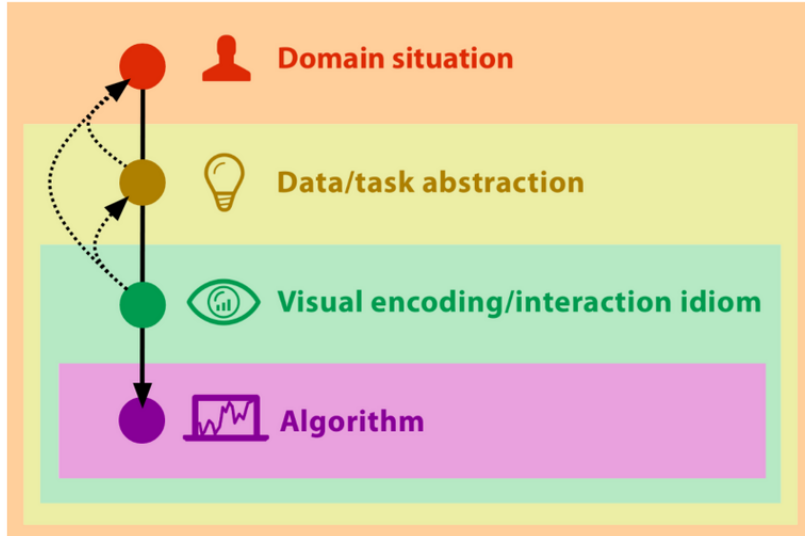
Participatory Visualization Design



technique-driven development

- development of new idioms, so that existing abstractions are better supported
- development of a new algorithm
- comparison with competing methods necessary

Participatory Visualization Design

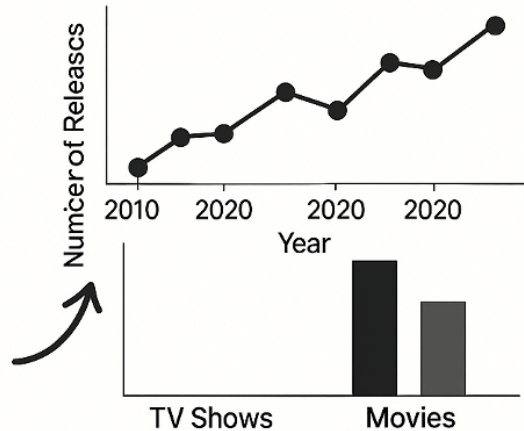


When it does not work...

Task: What marks and channels could you use to visualize this dataset?

show_id		type	director	country
s1	1	Breaking Bad	Vince Gilligan	United States
s2	2	The Irishman	Martin Scorsese	Unites States
s3	3	Parasite	Bong JHo	South Korea
s4	T	Stranger Things	The Duffer Brothers	United States
s5	5	Roma	Alfonso Cuaron	Mexico
s6	6	Darkest Hour	Joe Wright	United Kingdon

NETFLIX



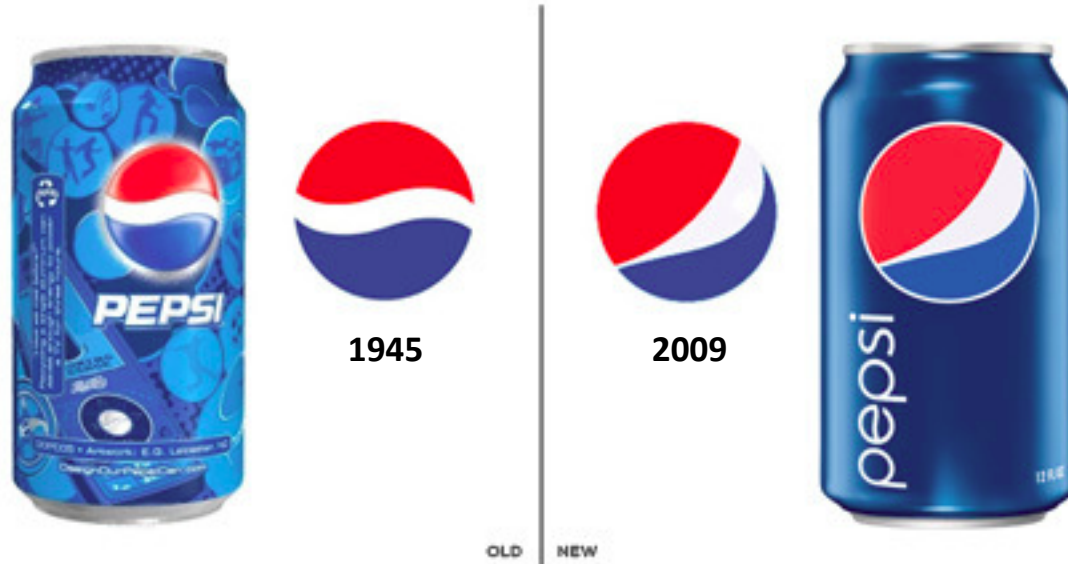
Data Analysis and Visualization

Hands-on: Miroboard!

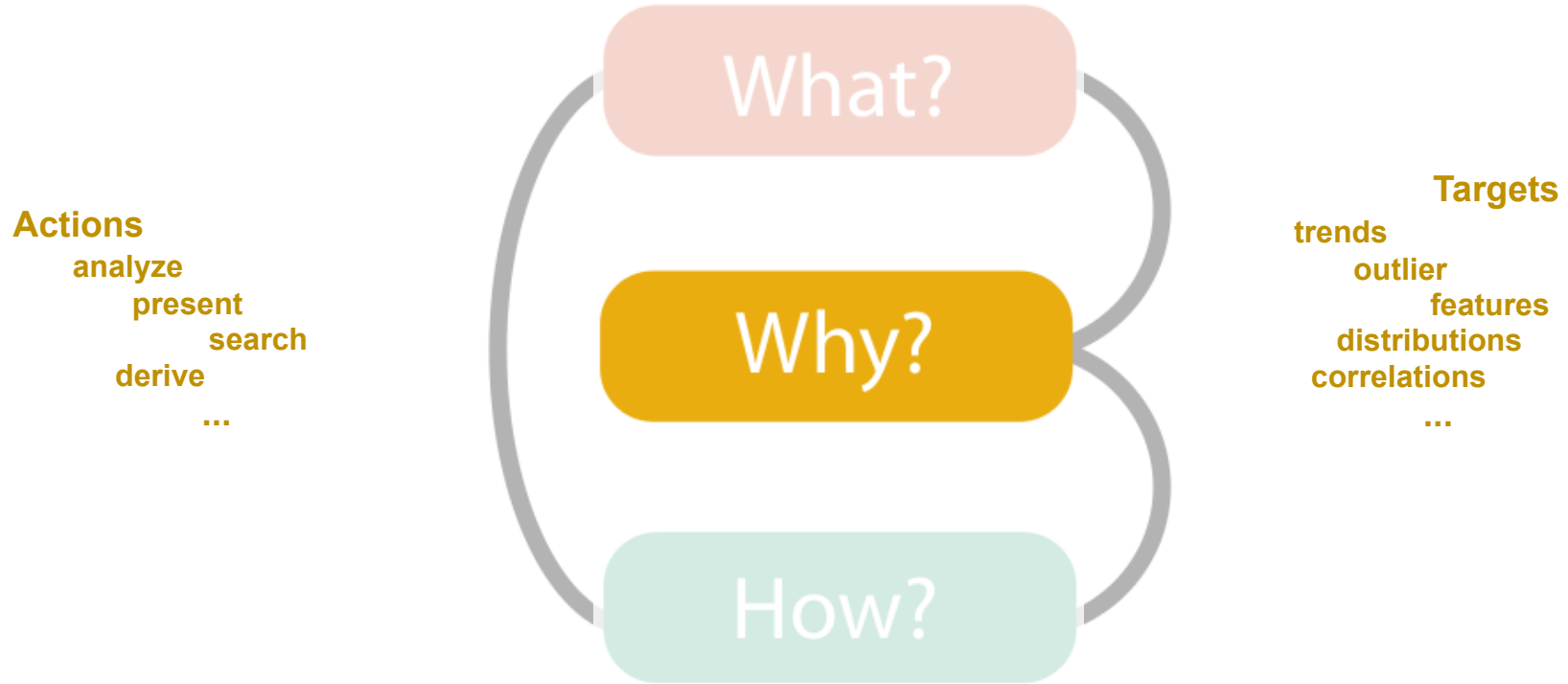
<https://tinyurl.com/56xx4ct5>



Task: Can you imagine why there was a shitstorm after the new Pepsi logo was released?



Participatory Visualization Design



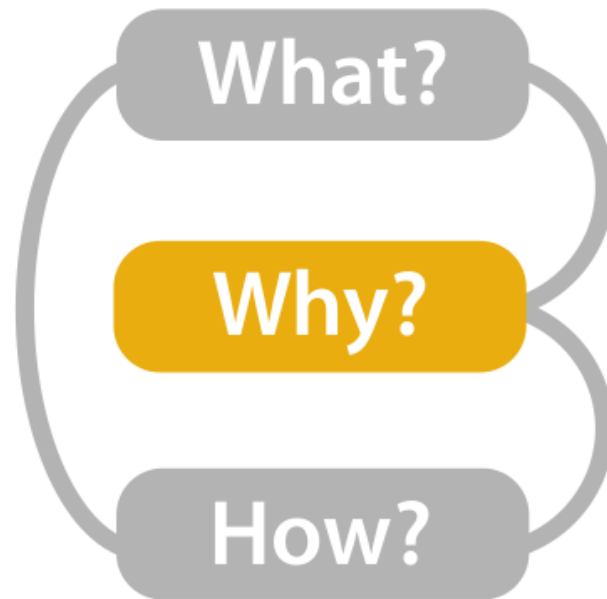
Task Abstraction

Does the research question need a visualization to be investigated?

Which tasks need to be supported by the visualization?

Can we map domain-specific to abstract research tasks?

What actions are directed towards which targets?



Why do people use a visualization?

actions
define
verbs

Actions

→ Analyze

→ Consume

→ Discover

→ Present

→ Enjoy



→ Produce

→ Annotate

→ Record

→ Derive



→ Search

	Target known	Target unknown
Location known	••• Lookup	••• Browse
Location unknown	<•••> Locate	<•••> Explore

→ Query

→ Identify

→ Compare

→ Summarize



Targets

→ All Data

→ Trends



→ Outliers



→ Features



→ Attributes

→ One

→ Distribution



→ Extremes



→ Many

→ Dependency



→ Correlation

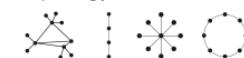


→ Similarity



→ Network Data

→ Topology



→ Paths



→ Spatial Data

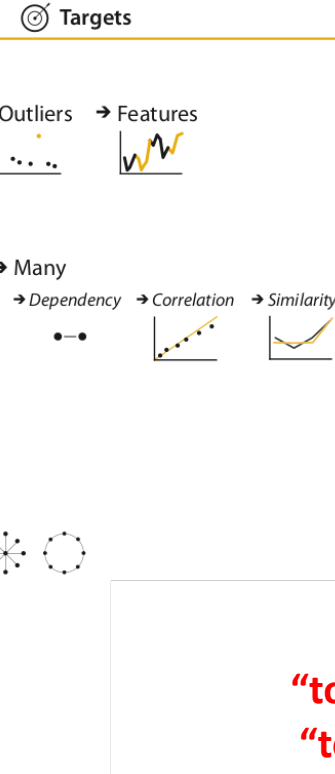
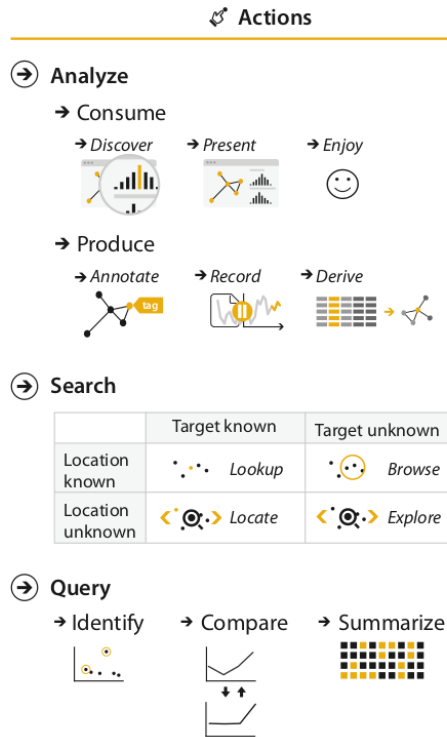
→ Shape



targets
define
nouns

Why do people use a visualization?

actions
define
verbs



targets
define
nouns

“to explore topologies”
“to identify extremes”
“to compare trends”

Actions

→ Analyze

→ Consume (*existing information*)

→ Discover



→ Present



→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



→ Derive



Actions

→ Analyze

→ Consume (*existing information*)



→ Present



→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



→ Derive



Discover

- find unknown information
- generate hypotheses
- verify existing hypotheses
- designer is often not aware of what the user wants to see (a reason for interaction means)

SnakeTrees: Discover new relationships

☐ MultiLine view ☒ SnakeLine view

☐ Point selection (Lasso) ☒ Cell selection (Click)

DR-Method

tsne

RESET SELECTION

Start Year

End Year

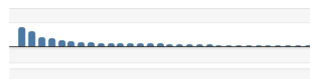
Location

— Artist Name

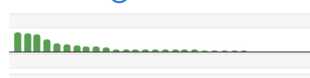
Quincy Jones

FETCH DATA

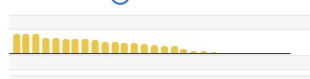
Artist: Quincy Jones

genre: 

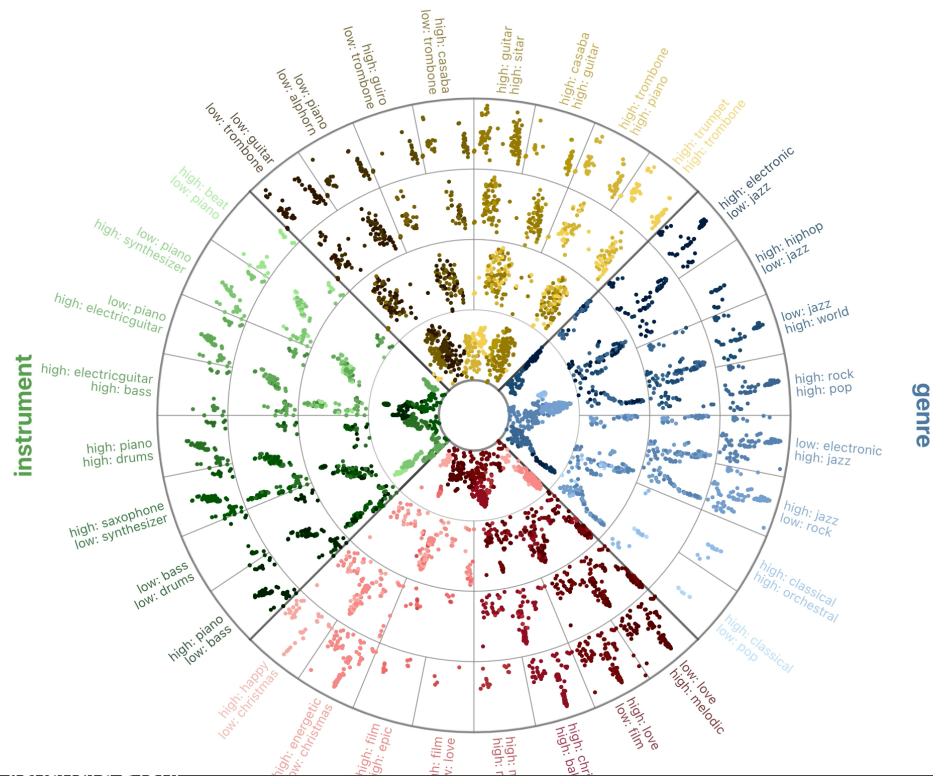
mood: 

instrument: 

inst_family:



Datapoint selection



Actions

→ Analyze

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→ Present



→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



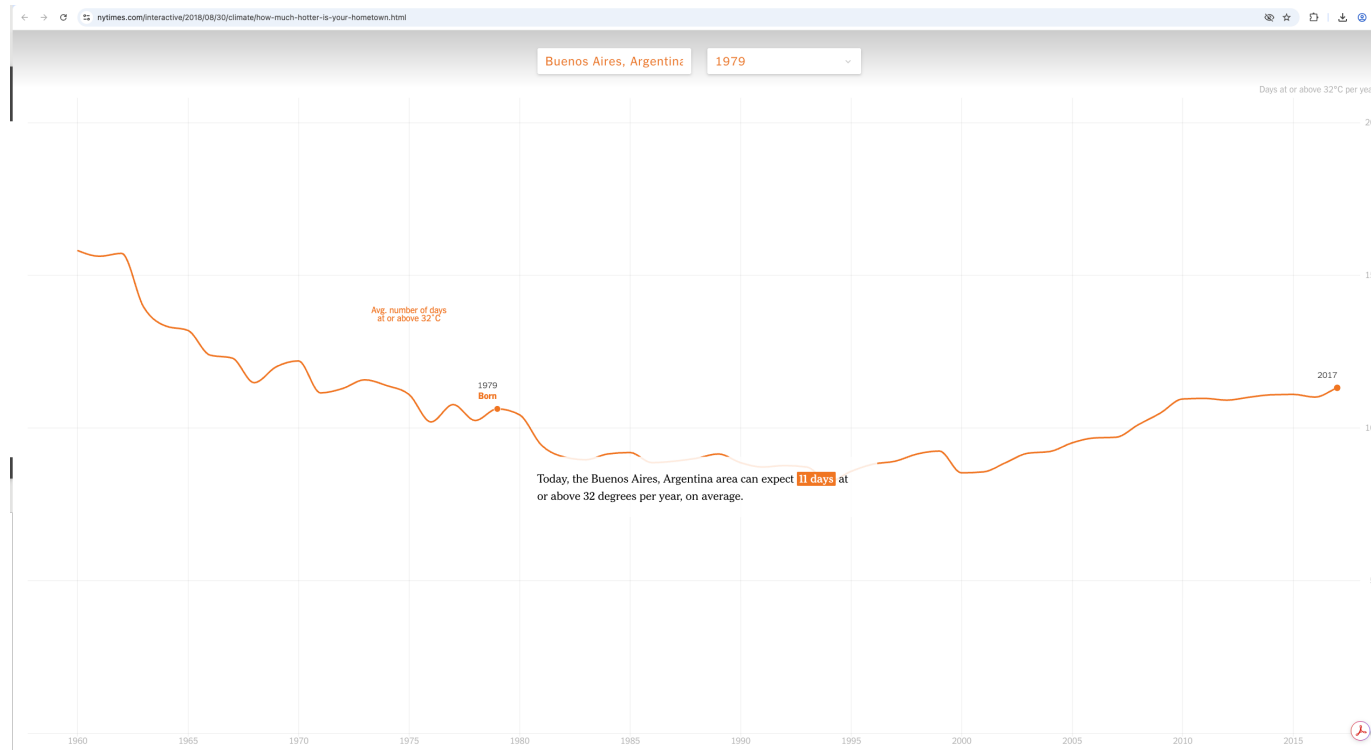
→ Derive



Present

- present information with visualization
- support decision making, planning, prediction, teaching
- existing concepts and facts are communicated by presenter
- visualization for illustration

Example: How Much Hotter is Your Hometown than When You were Born



<https://www.nytimes.com/interactive/2018/08/30/climate/how-much-hotter-is-your-hometown.html>

Actions

→ Analyze

→ Consume (*existing information*)

→ Discover



→ Present



→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



→ Derive

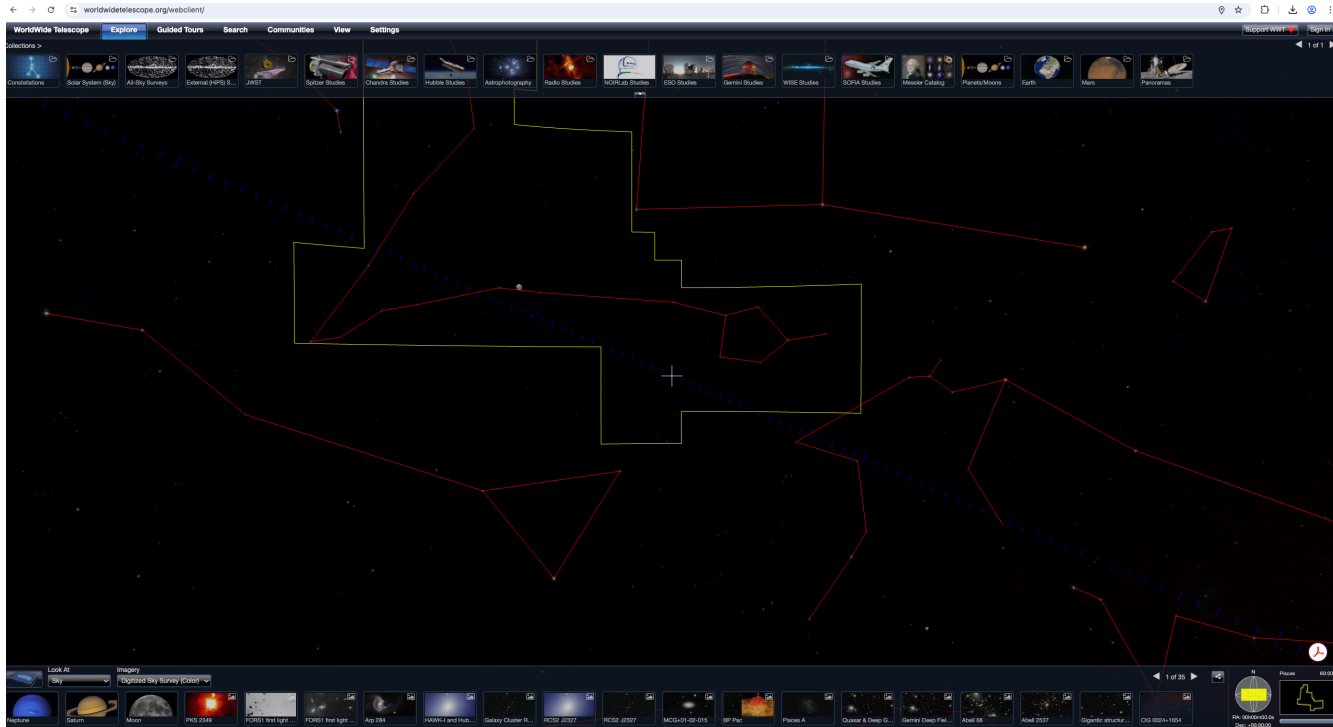


Enjoy

- casual consumption of a visualization without fixed goals or existing needs
- e.g. infographics, visualizations in museums

→ phenomenon: visualizations designed for a specific purpose may be used for casual exploration

Enjoy: World Wide Telescope



<https://worldwidetelescope.org/webclient/>

→ Analyze

→ Consume (*existing information*)

→ Discover



→ Present

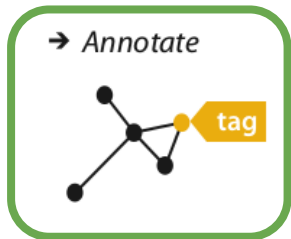


→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



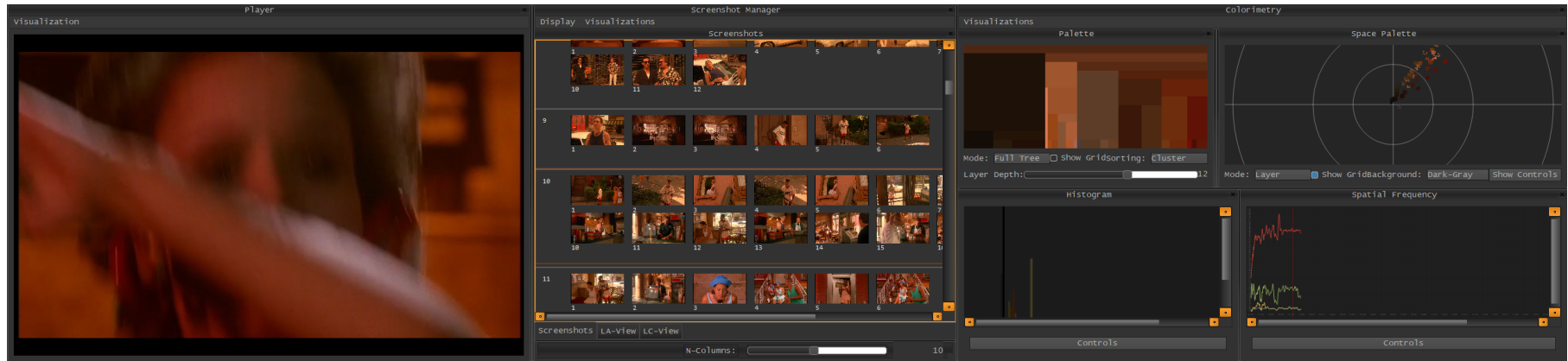
→ Derive



Annotate

- generating new information based on visual input
- adding new attributes to items (often manually), e.g., adding text categories to words in a text

VIAN: Video Annotation and Analysis



Actions

→ Analyze

→ Consume (*existing information*)

→ Discover



→ Present



→ Enjoy



→ Produce (*new information*)

→ Annotate



→ Record



→ Derive



Record

– saving visual elements as persistent artifacts

→ e.g. screenshots, creating a graphical history (all steps at of the use of a work are recorded)

→ allows users to access previous configurations/parameter settings

Actions

→ Analyze (*existing information*)

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce (*new information*)

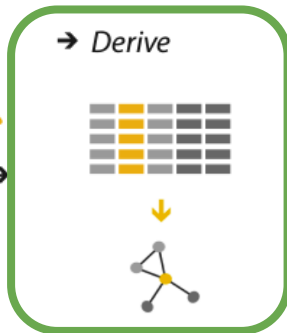
→ Annotate



→ Record



→ Derive



Derive

– create new data elements based on existing ones

– extend data with derived attributes that meet the needs of the user

→ for example: adding geo-references, recognition of relations

➔ Search (for data elements)

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

- *lookup*: Find a known element at a known position
- *locate*: Find a known element at an unknown position
- *browse*: Finding an unknown element at a known position
- *explore*: search for special features, starting with overview

➔ Search (for data elements)

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
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- *lookup*: Find a known element at a known position
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- *explore*: search for special features, starting with overview

→ Query

→ Identify



→ Compare



→ Summarize



- *identify*: examine the properties of a specific data element or identification of a specific data element with certain properties
- *compare*: compare multiple data elements
- *summarize*: create an overview of data

actions relate to targets → aspects of data of interest to the user

→ **All Data** (*high-level targets*)

→ Trends



find patterns
in the data

→ Outliers



examine elements
that do not match
the pattern

→ Features



data- and task-
specific detection
of structures

➔ **Attributes** (*low-level targets on multivariate data*)

➔ One

➔ *Distribution*



➔ *Extremes*

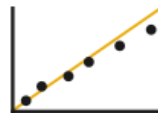


➔ Many

➔ *Dependency*



➔ *Correlation*



➔ *Similarity*



- finding a value for an attribute
- finding extreme values
- examine the distribution of values of a dimension

➔ Attributes *(low-level targets on multivariate data)*

➔ One

➔ Distribution



➔ Extremes

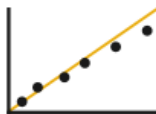


➔ Many

➔ Dependency



➔ Correlation



➔ Similarity



- values of one attribute are dependent on other attributes
- examine the correlation between attributes
- analysis of similarity between attribute values
- ➔ *Understanding trends, outliers, distributions and the correlation between attributes are common reasons for visualizations!*

Targets

➔ Network Data

➔ Topology



➔ Paths



- understand structure/topology of the network
- analyze paths between data elements

➔ Spatial Data

➔ Shape



- compare geometric shapes

Munzner, T. (2014). *Visualization analysis and design*. CRC press.

Chapter 3: Why: Task Abstraction

Brehmer, M., & Munzner, T. (2013). A multi-level typology of abstract visualization tasks. *IEEE transactions on visualization and computer graphics*, 19(12), 2376–2385.

Schulz, H. J., Nocke, T., Heitzler, M., & Schumann, H. (2013). A design space of visualization tasks. *IEEE Transactions on Visualization and Computer Graphics*, 19(12), 2366–2375.

Tasks on graphs / focus on relational data:

Lee, B. et al. (2006). Task taxonomy for graph visualization. In *Proceedings of the 2006 AVI workshop on BEyond time and errors: novel evaluation methods for information visualization* (pp. 1–5).